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WIND TURBINE WITH TWO SETS OF BLADES AND AEROGENERATOR

Abstract

The wind turbine comprises a rotor which rotates with respect to a stator about a rotation axis. The rotor comprises at least one narrow blade and at least one wide blade fixed with respect to one another, and the lateral extension of the wide blade is at least 1.5 times that of the narrow blade.

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Background/Summary

FIELD OF INVENTION

[0001] The present invention relates to the field of wind turbines.

BACKGROUND OF THE INVENTION

[0002] Wind turbines typically comprise an electrical generator which comprises a generator rotor rotating with respect to a generator stator about a rotation axis, this movement generating electricity. The rotor typically has one or more blades which are designed to catch the wind, so that the wind imparts the rotary motion to the rotor. In the field of wind turbines, one approach is to generate more and more power by ever-larger wind turbines. One issue however is that these very large wind turbines trouble human life and, therefore, they are to be placed far away from humans and human activities. This technology therefore relies on electrical cables to bring the generated electricity from where it is produced to where it is needed, but much energy is lost during transportation.

[0003] Recently, it was proposed in US 2015/108762 to use smaller such wind turbines, which can therefore be integrated where energy is needed. In order to make such wind turbines acceptable, a biomorphic approach was used, and many small wind turbines are provided on a tree-like structure.

[0004] The challenges associated with this technology are to generate enough power while not degrading the biomorphism, in order to ensure the acceptability.

SUMMARY OF THE INVENTION

[0005] According to the invention, it is provided a wind turbine, wherein the wind turbine comprises a rotor, wherein the rotor is designed to rotate with respect to a stator about a rotation axis, wherein the rotor comprises at least one narrow blade with a blade edge, wherein the at least one narrow blade has a first maximal lateral extension measured from the rotation axis to the blade edge of said narrow blade, [0006] and wherein the rotor comprises at least one wide blade with a blade edge, wherein the at least one wide blade has a second maximal lateral extension measured from the rotation axis to the blade edge of said wide blade, [0007] wherein the blades are fixed with respect to one another, [0008] wherein the second maximal lateral extension is at least 1.5 times the first maximal lateral extension.

[0009] Thanks to these features, while the biomorphic approach is kept, the starting threshold of the wind turbine is lowered, while not impairing the production of electricity in case of winds of higher speed. The starting threshold is the speed of the wind under which the wind does not cause the start of rotation of the wind turbine due to its inertia.

[0010] According to a specific embodiment, the narrow blade has a first maximal height along the rotation axis, and the wide blade has a second maximal height along the rotation axis, and the first maximal height is greater than the second maximal height.

[0011] According to a specific embodiment, the wind turbine comprises a narrow blade portion and a wide blade portion which are separated from one another along the rotation axis, and all narrow blades are provided in the narrow blade portion and all wide blades are provided in the wide blade portion.

[0012] According to a specific embodiment, the narrow blade portion is a single piece. This enables to limit or reduce the number of mechanical fastenings used in the narrow blade portion, which in turn is benefic to withstand dynamic stresses and fatigue which typically occur at fastenings in mechanical devices. Since less energy is dissipated at fastenings, more of the imparted energy may contribute to the production of electricity.

[0013] According to a specific embodiment, the narrow blade portion is made of an additive manufacturing material.

[0014] According to a specific embodiment, the at least one narrow blade has a thickness which continuously increases from the blade edge.

[0015] According to a specific embodiment, the wide blade portion is mechanically assembled to the narrow blade portion.

[0016] According to a specific embodiment, the wide blade portion comprises a plurality of blade components which are assembled to one another.
[0017] According to a specific embodiment, the blade components are identical to one another. This enables to reduce the tooling necessary for production, which has a positive energetic impact.
[0018] According to a specific embodiment, the wind turbine further comprises an electrical generator comprising a generator rotor and a generator stator, wherein the generator rotor is designed to rotate with respect to the generator stator about said rotation axis, wherein the generator rotor is assembled to the rotor of the wind turbine, wherein the electrical generator is housed within the wide blade portion.
[0019] According to a specific embodiment, the wide blade comprises a plurality of lobes separated from one another by a recess.
[0020] According to a specific embodiment, the wind turbine further comprises an electrical generator comprising a generator rotor and a generator stator, wherein the generator rotor is designed to rotate with respect to the generator stator about said rotation axis, wherein the generator rotor is assembled to the rotor of the wind turbine.
[0021] According to a specific embodiment, the wind turbine comprises a first number of at least two narrow blades, wherein the narrow blades are equally spaced from one another about the rotation axis, wherein the wind turbine comprises a second number of at least two wide blades, wherein the wide blades are equally spaced from one another about the rotation axis.
[0022] According to a specific embodiment, the first number is strictly greater than the second number.
[0023] According to another aspect, the invention relates to an aerogenerator comprising a structure and a plurality of such individual wind turbines, wherein electrical current generated by the aerogenerator is a sum of electrical currents generated by said wind turbines.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0024] FIG. **1** is a three-dimensional top perspective view of a wind turbine according to one embodiment.
[0025] FIG. **2** is an exploded bottom perspective view of the wind turbine of FIG. **1**.
[0026] FIG. **3** is a top sectional view of the narrow blade portion of the wind turbine of FIG. **1**.
[0027] FIG. **4** is a front sectional view of the narrow blade portion of the wind turbine of FIG. **1**.
[0028] FIG. **5** is a top view of the wide blade portion of the wind turbine of FIG. **1**.
[0029] FIG. **6** is a partial front sectional view of the wind turbine of FIG. **1**.
[0030] FIG. **7** is a perspective view of a first embodiment of an aerogenerator.
[0031] FIG. **8** is a perspective view of a second embodiment of an aerogenerator.
DETAILED DESCRIPTION OF THE INVENTION
[0032] In the present invention where gravity plays a role, “vertical” is used to designate the direction passing through the center of the Earth. “Horizontal” is used to designate the plane normal to the “vertical” direction. Words like “up”, “down”, “high”, “low”, “above”, “below”, etc. are used with respect to the vertical direction.
[0033] The present description is provided assuming the ground is horizontal. The invention would however be applicable everywhere, even if the ground is not strictly speaking horizontal. Wind is a flow of air which is typically parallel to the horizontal. The invention remains applicable when wind flows not strictly speaking horizontally, for example due to the shape of the ground, due to obstacles to flow such as trees, buildings or else, or for other reasons.
[0034] FIG. **1** schematically represents an embodiment of a wind turbine **1** according to one embodiment of the invention. A wind turbine **1** comprises a stator **2** and a rotor **3**. The rotor **3** is

mounted to rotate with respect to the stator **2** about a fixed rotation axis (Z.sub.r). The present invention is applicable to so-called “vertical axis” wind turbines. A “vertical axis” wind turbine is a wind turbine designed to be used with its rotation axis Z.sub.r extending along the vertical direction. In the following description, it will be assumed that the wind turbine **1** is placed so that its rotation axis Zr extends vertically. Of course, the same turbine may be oriented differently.
[0035] The rotation axis is central for the rotor **3**.
[0036] As can be seen on FIG. **1**, the rotor **3** comprises a narrow blade portion **4** and a wide blade portion **5**. The narrow blade portion **4** and the wide blade portion **5** are spaced from one another along the vertical axis. In the presented embodiment, the narrow blade portion **4** is located above the wide blade portion **5**.
[0037] The narrow blade portion **4** is narrower than the wide blade portion **5**. The breadth of the blade portions is measured as the maximal horizontal distance between the rotation axis and any point of the respective blade portion.
[0038] As can also be seen on FIG. **1**, the narrow blade portion **4** is higher than the wide blade portion **5**. The height of the blade portions is measured as the maximal vertical distance between any two points of the respective blade portion.
[0039] The narrow blade portion has a plurality of blades **6**. The blades of the narrow blade portion are called “narrow blades” **6**. These narrow blades **6** are equally distributed about the rotation axis. Further, the narrow blades **6** are identical to one another. This ensures a homogeneous transformation of wind energy to electrical energy. In the present example, there are three identical narrow blades **6** which form angles of 120° two by two about the rotation axis. Therefore, below, the description of a single narrow blade **6** will be provided, and this description will be applicable to the other narrow blades.
[0040] Below, we will describe one horizontal section of a narrow blade **6**, with reference to FIG. **3**. The rotor is designed to rotate in the clockwise direction on FIG. **3**. The section comprises a proximal portion **7** and a distal portion **8**. The words “proximal” and “distal” are used by reference to the central rotation axis.
[0041] The narrow blade **6** is a thin shell shaped to receive incoming wind. The shape comprises a curvature in the horizontal plane, as can be seen in FIG. **3**. As can be seen on the other figures, the shape may also comprise a curvature in a vertically secant plane. So, overall, the wide blade **18** defines a cavity **50**. The narrow blade **6** comprises a concave front face **9** and an opposed convex back face **10**. The front and back faces **9** and **10** are sensibly parallel to one another. The thickness of the narrow blade **6** is sensibly constant, from the proximal portion **7** to the distal portion **8**. This thickness, also called “main thickness” is for example comprised between 1.5 millimeters (mm) and 5 mm, preferably between 2.5 and 3.5 mm. This thickness of a suitable material allows supporting the imparting energy of wind with a minimum inertia. The distal portion **8** may have a continuously decreasing thickness from the connection to the proximal portion **7** to the edge **11**. The thickness at the edge, also called “edge thickness”, is for example not lower than 0.5 mm, notably not lower than 1.5 mm. This thickness allows sufficient strength of the edge. The ratio of the main thickness to the edge thickness is for example between 1.5 and 10, for example between 2.5 and 3.5. This allows manufacturability, strength, low inertia and low drag coefficient Cx of the blade.
[0042] As can be seen also from FIG. **3**, the rotation axis Zr is disposed sensibly between the proximal portion **7** and the distal portion **8**.
[0043] Along the vertical axis, the narrow blade **6** extends from a bottom portion **12** to a top portion **13**. The bottom portion **12** is connected to a base **14** along its whole width. The top portion **13** is connected to a top **15** along its whole width. These two connections ensure a firm holding of the narrow blade **9** within the rotor **3**.
[0044] Along the vertical axis, the edge **11** is also continuously curved. The location with the maximal lateral extension is between the bottom portion **12** and the top portion **13**. More precisely,

this location is closer to the bottom portion **12** than to the top portion **13**, measured along the vertical axis. This enables the narrow blade portion **4** of the rotor **3** to look like a tree leaf. For example, the distance between the bottom portion **12** and the location with the maximal lateral extension of the narrow blade **6**, along the vertical axis, is between ⅕ and ⅓ of the height of the narrow blade **6**, along the vertical axis. The tapering shape of the narrow blade portion **4** allows to reduce the moments of force applied to the assembly of the wind turbine to its support.
[0045] The top **15** may comprise a continuous structure **16** which is connected to the top portions **13** of all narrow blades **6**, for example a flat plate. The top **15** may further comprise an outer dome **17** which is connected to the continuous structure **16** and overlies it. The dome **17** strengthens the rotor **3** and provides a rounded top shape which is harmless in case of contacts and enables the rotor to look like a tree leaf.
[0046] The base **14** may be conically shaped pointing upward, with the rotation axis Zr as cone axis. This shape enables to provide space to house other components, such as will be described below. Further, this geometry strengthens the connection of the narrow blade portion **4** with the underlying elements which will be described below.
[0047] According to one embodiment, the narrow blade portion **4** which has been described above is single-piece. This enables to get rid of mechanical connection devices which allow play between components, and are subject to vibration. For example, the narrow blade portion **4** is manufactured by additive manufacturing. This additive manufacturing technology allows to manufacture a single piece with such a complex shape as that of the narrow blade portion **4**. In such case, the narrow blade portion **4** is made of a suitable additive-manufacturing material, such as polylactic acid (“PLA”) or acrylonitrile butadiene styrene (“ABS”) or the like. One may choose a biodegradable, non-toxic, rigid, strong and/or sustainable material. So, when, in the description above, it is mentioned that two parts, sections or portions, are “connected”, it should be understood that they are formed of an integral single-piece of material under this embodiment. Additive-manufacturing enables to obtain a thin blade with good mechanical strength, which lowers weight, inertia and cost of manufacturing with respect to thicker blades.
[0048] In addition, by reducing the mechanical fastenings, less energy is dissipated as mechanical energy at such fastenings, and the risk of loosening of attachments is reduced. Loosened attachments may cause dynamic stresses and fatigue to the wind turbine. Providing the narrow blade portion **4** as a single piece thus enables to increase the life expectancy of the wind turbine **1**, and to maintain a high ratio of energy conversion along time.
[0049] As can be seen on FIG. **5**, below, the rotor **3** further comprises the wide blade portion **5**.
[0050] The wide blade portion **5** is wider than the narrow blade portion **4**. For example, the maximal lateral extension of the wide blade portion **5** is at least 1.5 times the maximal lateral extension of the narrow blade portion **4**. Notably, the ratio of the maximal lateral extension of the wide blade portion **5** to the maximal lateral extension of the narrow blade portion **4** is at least equal to 1.6, at least equal to 1.7, at least equal to 1.8, at least equal to 1.9, or at least equal to 2.
[0051] According to some embodiments, the ratio of the maximal lateral extension of the wide blade portion **5** to the maximal lateral extension of the narrow blade portion **4** is less than 2.5, notably less than 2.4, less than 2.3, less than 2.2, or even less than 2.
[0052] According to some embodiments, the diameter of the circle circumventing the wide blade portion **5**, and hence the wind turbine **1**, is comprised between 0.5 meters (m) and 1 m, preferably between 0.65 m and 0.7 m.
[0053] As can also be seen on FIG. **1**, the wide blade portion **5** is shorter than the narrow blade portion **4**, as measured along the vertical axis. For example, the maximal height of the narrow blade portion **4** is at least 1.5 times the maximal height of the wide blade portion **5**. Notably, the ratio of the maximal height of the narrow blade portion **4** to the maximal height of the wide blade portion **5** is at least equal to 1.6, at least equal to 1.7, at least equal to 1.8, at least equal to 1.9, or at least equal to 2.

[0054] According to some embodiments, the ratio of the maximal height of the narrow blade portion **4** to the maximal height of the wide blade portion **5** is less than 3.5, notably less than 3 or even less than 2.5.
[0055] According to some embodiments, the height of the wind turbine **1** along the vertical axis, is comprised between 0.7 m and 1.5 m, preferably between 1.0 m and 1.2 m.
[0056] The wide blade portion **5** has a plurality of blades **18**. The blades of the wide blade portion **5** are called “wide blades” **18**. These wide blades **18** are equally distributed about the rotation axis. Further, the wide blades **18** are identical to one another. This ensures a homogeneous transformation of wind energy to electrical energy.
[0057] According to one embodiment, the number of wide blades **18** is identical to the number of narrow blades **6**. In such an embodiment, narrow blades **6** and wide blades **18** may be aligned with one another two by two. According to a variant, the wide blades **18** are rotationally offset with respect to the narrow blades **6** about the rotation axis. The offset is for example half of the angle between two following narrow blades **6**.
[0058] According to another embodiment, the number of wide blades **18** is different from the number of narrow blades **6**. For example, the difference between the number of wide blades **18** and the number of narrow blades **6** is equal to 1. According to another example, it is equal to 2. For example, it is less than 6, notably less than 4.
[0059] In the present example, there are two identical wide blades **18** which form an angle of 180° with one another about the rotation axis. In addition, no wide blade **18** is aligned with a corresponding narrow blade along the vertical axis. The angle between a narrow blade **6** and a wide blade **18** is at least 30°.
[0060] Below, the description of a single wide blade **18** will be provided, and this description will be applicable to the other wide blade(s) **18**.
[0061] Below, we will describe one wide blade **18**, with reference to FIG. **5**. The wide blade **18** comprises a proximal portion **19** and a distal portion **20**. The words “proximal” and “distal” are used by reference to the central rotation axis.
[0062] The wide blade **18** is a thin shell shaped to receive incoming wind. The shape comprises a curvature in the horizontal plane, as can be seen in FIG. **5**. As can be seen on the other figures, the shape may also comprise a curvature in a vertically secant plane. The wide blade **18** comprises a concave front face **21** and an opposed convex back face **22**. With respect to the rotation axis, the concavity of the wide blade **18** is the same as that of narrow blade **6**. The front and back faces **21** and **22** are sensibly parallel to one another. The thickness of the wide blade **18** is sensibly constant, from the proximal portion **19** to the distal portion **20**. For example, this thickness, also called “main thickness” is the same as that of the narrow blade **6**. This thickness allows supporting the imparting energy of wind with a minimum inertia. The distal portion **20** may have a continuously decreasing thickness from the connection to the proximal portion **19** to the edge **23**. In other words, the thickness continuously increases, starting from the edge. The thickness at the edge, also called “edge thickness”, is for example not lower than 0.5 mm. For example, it is the same as for the narrow blade **6**. This thickness allows sufficient strength of the edge. The ratio of the main thickness to the edge thickness is for example between 1.5 and 10. For example, it is the same as for the narrow blade **6**. This allows manufacturability, strength, low inertia and low drag coefficient Cx of the blade.
[0063] According to one embodiment, as shown in particular on FIGS. **1** and **6**, the wide blade **18** comprises at least two lobes **40**. The two lobes **40** are spaced from one another along the rotation axis Zr. In other words, the wide blade **18** comprises a top lobe **40***a* and a bottom lobe **40***b*. Two consecutive lobes **40***a*, **40***b* along the rotation axis are therefore separated by a recess **41**. This configuration allows to reduce the drag coefficient of the wide blade **18**.
[0064] According to one embodiment, the wide blade portion **5** comprises a bottom housing **24** from which the wide blades **18** extend. The bottom housing **24** is for example cup-shaped, with a

narrow bottom and a large upper opening **26**. The cup is for example partly spherical, or partly cylindrical of revolution. The bottom housing **24** has a shell **27** from which the wide blades **18** extend radially outward. The wide blade **18**, and notably the top lobe **40***a* extends above the top plane of the bottom housing **24**.

[0065] For example, the wide blade portion is made of a plurality of blade components which are assembled to one another by any suitable means, notably screwing, bolting, crimping or snap-fastening. According to one example, all blade components are identical, which enables to reduce the number of molds necessary for the production of the wide blade. Each blade component may comprise one single wide blade **18** and a portion of the bottom housing **24**. For example, as shown on FIG. **5**, the wide blade portion is made of two identical blade components **28***a* and **28***b* which are assembled to one another. Reference "**28**" is used to generally designate one of these blade components.

[0066] According to one embodiment, each of the blade components **28** which has been described above is single-piece. This enables to get rid of mechanical connection devices which allow play between components, and are subject to vibration. For example, the blade component **28** is manufactured by additive manufacturing. This additive manufacturing technology allows to manufacture a single piece with such a complex shape as that of the blade component **28**. In such case, the blade components **28** is made of a suitable additive-manufacturing material, such as polylactic acid ("PLA") or acrylonitrile butadiene styrene ("ABS") or the like. One may chose a biodegradable, non-toxic, rigid, strong and/or sustainable material. Possibly, one will use the same material as that of the narrow blade portion **4**, in order to obtain similar mechanical properties. So, when, in the description above, it is mentioned that two parts, sections or portions, are "connected", it should be understood that they are formed of an integral single-piece of material under this embodiment. According to an embodiment, the material constituting the blade component **28** is the same as that constituting the narrow blade portion **4**.

[0067] The narrow blade portion **4** and the wide blade portion **5** are fixedly assembled to one another by any suitable means, such as screwing, bolting, crimping, snap-fastening, or the like. Preferably, the assembly may be undone using suitable tools, for example for maintenance.

[0068] The wind turbine **1** further comprises an electrical generator **29**, such as shown on FIG. **6**. The electrical generator **29** comprises a generator rotor **30** and a generator stator **31**. The generator rotor **30** is designed to rotate with respect to the generator stator **31** about a rotation axis. When the electrical generator **29** is assembled within the wind turbine, this rotation axis is the rotation axis Zr. The generator rotor **30** is assembled to the rotor **3** of the wind turbine **1**. In particular, as shown on FIG. **6**, the electrical generator **29** is housed within the wide blade portion **5**, notably in the bottom housing **24**, and attached thereto and/or to the base **14** of the narrow blade portion **4** by any suitable means.

[0069] According to one embodiment, the generator rotor **30** may comprise a plurality of permanent magnets with alternated polarity, and the generator stator **31** has electrical coils intended to interact with the permanent magnets of the generator rotor **30**, so that rotation of the rotor **3** generates an alternating electrical field at the generator stator **31**.

[0070] The wind turbine **1** further comprises an electronical circuit **32**. The electronical circuit **32** comprises a microprocessor **39** adapted to regulate the electrical current outputted from the wind turbine **1**. For example, the electrical current generated by the wind turbine **1** is a 3-phase 48V alternating current.

[0071] According to some embodiments, a redresser (not shown) of the electronical circuit **32** may redress the alternating current to a DC current.

[0072] A comparison example is a wind turbine as described above, with the same blade surface as the invention, but with identical widths in the bottom and top blade portions. So, compared with the wind turbine of the invention, the width of the bottom blade portion of the comparison example is less than the width of the wide blade portion of the invention, and the width of the top blade

portion of the comparison example is greater than the width of the narrow blade portion of the invention.
[0073] Assuming a starting condition of low wind, so that the wind turbine **1** is immobile, when wind speed rises, the wide blade portion enables a lower starting threshold for the wind turbine **1** of the invention as compared to the comparative example. This is due to the fact that the forces exerted at the edge of the wide blades **18** exhibit a larger moment with respect to the rotation axis. So, compared with the comparative example, the wind turbine **1** produces electrical energy at slow winds, where the comparative example does not. As wind picks up, it imparts rotation to the narrow blades **6**, which have a lower moment of inertia, which increases the speed of rotation and hence the intensity of generated electrical current.
[0074] According to some embodiments, wind turbines such as described above have a starting threshold of less than 10 m/s, even less than 6 m/s, preferably less than 5 m/s.
[0075] So, overall, the time during which the wind turbine **1** according to the invention produces electricity is greater than that of the comparative example.
[0076] As can be seen on FIG. **7**, according to one embodiment, an aerogenerator **33** comprises a plurality of wind turbines. For example, one, at least one, more than one or all of the wind turbines are wind turbines **1** according to the above description.
[0077] For example, the electrical generators **29** of one or more or all of the wind turbines of the aerogenerator **33** are each controlled independently from one another by their respective microcontroller **39**. They are electrically connected to one another in parallel, so that the electrical current output by the aerogenerator **33** is the sum of the electrical currents generated by the individual wind turbines **1**. This allows for example various wind turbines to rotate at different speeds (or even potentially some of them not rotating), due to the local flow of wind, screening effects, or any other reasons, and to adapt to ever-changing wind conditions.
[0078] The microcontroller **39** may implement an AI-based rule for the control of its individual wind turbine.
[0079] As shown on FIG. **7**, the aerogenerator **33** comprises a structure **34** which carries the wind turbines **1**. The structure **34** is designed to place the carried wind turbines **1** in space according to an arrangement which maximizes electrical energy output for incoming wind. For example, wind turbines are arranged so that screening between wind turbines is null or low. Screening relates to a part of a wind turbine being behind a part of another wind turbine along a direction of wind flow. Further, the wind turbines are arranged so that, taking into account a cylinder circumscribing the wind turbines together, along a direction of wind flow, the cross-section of this cylinder which is not intercepted by a wind turbine is minimal.
[0080] For example, the wind turbines **1** are offset from one another in all three directions of space.
[0081] The structure **34** may for example comprise a foot **35** and a plurality of poles **36** extending sensibly vertically from the foot **35**, and a wind turbine **1** assembled to each pole **36**. The wind turbine **1** may be assembled to a pole **36** by any suitable means. In particular, the generator stator **31** is assembled to the pole **36**. Electrical wires (not shown), which deliver electrical energy from the wind turbine **1** may extend inside the inner lumen of the pole **36** to the foot **35**, which houses an electrical cabinet.
[0082] According to another embodiment, as shown on FIG. **9**, the aerogenerator **33** may comprise a structure **34** with a more complex shape, comprising a common trunk **37** from which extend, at various levels, and in various directions, branches **38** which each carry one or more wind turbines **1**.
REFERENCES
[0083] Wind turbine **1** [0084] Stator **2** [0085] Rotor **3** [0086] Narrow blade portion **4** [0087] Wide blade portion **5** [0088] Narrow blade **6** [0089] Proximal portion **7** [0090] Distal portion **8** [0091] Front face **9** [0092] Back face **10** [0093] Edge **11** [0094] Bottom portion **12** [0095] Top portion **13** [0096] Base **14** [0097] Top **15** [0098] Structure **16** [0099] Dome **17** [0100] Wide blade **18** [0101]

Proximal portion **19** [0102] Distal portion **20** [0103] concave front face **21** [0104] convex back face **22** [0105] edge **23** [0106] bottom housing **24** [0107] bottom **25** [0108] upper opening **26** [0109] shell **27** [0110] blade components **28***a*, **28***b* [0111] generator **29** [0112] generator rotor **30** [0113] generator stator **31** [0114] electronical circuit **32** [0115] aerogenerator **33** [0116] structure **34** [0117] foot **35** [0118] pole **36** [0119] trunk **37** [0120] branches **38** [0121] microprocessor **39** [0122] lobe **40**, **40***a*, **40***b* [0123] recess **41** [0124] cavity **50**

Claims

1. (canceled)
2. (canceled)
3. (canceled)
4. The wind turbine according to claim **18**, wherein said first blade portion is a single piece.
5. The wind turbine according to claim **18**, wherein said first blade portion is made of an additive manufacturing material comprising polylactic acid or acrylonitrile butadiene styrene.
6. (canceled)
7. (canceled)
8. The wind turbine according to claim **18**, wherein said plurality of second blades are fixed to one another.
9. (canceled)
10. The wind turbine according to claim **18**, further comprising an electrical generator comprising a generator rotor and a generator stator, wherein the generator rotor is designed to rotate with respect to the generator stator about said rotational axis, wherein the generator rotor is fixed to the rotor of the wind turbine, wherein the electrical generator is housed within the bottom housing of said second blade portion.
11. The wind turbine according to claim **18**, wherein each second blade comprises a plurality of lobes separated from one another by a recess.
12. The wind turbine according to claim **18**, further comprising an electrical generator comprising a generator rotor and a generator stator, wherein the generator rotor is designed to rotate with respect to the generator stator about said rotational axis, wherein the generator rotor is fixed to the rotor of the wind turbine.
13. (canceled)
14. The wind turbine according to claim **18**, having a number of first blades and a number of second blades, and wherein the number of first blades is greater than the number of second blades.
15. An aerogenerator, comprising: a plurality of individual wind turbines each according to claim **18**, and a structure for carrying said plurality of individual wind turbines, wherein electrical current generated by the aerogenerator is a sum of electrical currents generated by said wind turbines.
16. (canceled)
17. (canceled)
18. A wind turbine, comprising: a rotor, wherein the rotor is rotatable with respect to a stator about a vertically aligned rotational axis, wherein the vertical direction passes through a center of Earth, wherein the rotor comprises a first blade portion coaxially aligned with a second blade portion along the vertically aligned rotational axis, wherein the first blade portion comprises a base and a plurality of first blades extending from the base which are identical to one another and are equally distributed about the vertically aligned rotational axis, wherein each first blade extends from a bottom portion to a top portion and has a maximal vertical height along the vertically aligned rotational axis, wherein the rotation of the first blade portion about the vertically aligned rotational axis circumscribes a circle having a radius Wn, wherein said second blade portion comprises a bottom housing and a plurality of second blades extending from the bottom housing which are identical to one another and are equally distributed about the vertically aligned rotational axis,

wherein each second blade extends from a bottom portion to a top portion and has a maximal vertical height along the vertically aligned rotational axis, wherein the rotation of the second blade portion about the vertically aligned rotational axis circumscribes a circle having a radius Ww, and wherein the first blade portion and the second blade portion are fixed with respect to one another, the maximal vertical height for each first blade is at least 1.5 times greater than the maximal vertical height for each second blade, and the radius Ww for the second blade portion is at least 1.5 times greater than the radius Wn for the first blade portion.

19. The wind turbine according to claim 18, wherein the first blade portion and the second blade portion are fixed with respect to one another by the base of the first blade portion being fixed to the bottom housing of the second blade portion.

20. A wind turbine, comprising: a rotor, wherein the rotor is rotatable with respect to a stator about a vertically aligned rotational axis, wherein the vertical direction passes through a center of Earth, wherein the rotor comprises a first blade portion coaxially aligned with a second blade portion along the vertically aligned rotational axis, wherein the first blade portion comprises a base and a plurality of first blades extending from the base which are identical to one another and are equally distributed about the vertically aligned rotational axis, wherein each first blade extends from a bottom portion to a top portion and has a maximal vertical height along the vertically aligned rotational axis, wherein the rotation of the first blade portion about the vertically aligned rotational axis circumscribes a circle having a radius Wn, wherein said second blade portion comprises a bottom housing and a plurality of second blades extending from the bottom housing which are identical to one another and are equally distributed about the vertically aligned rotational axis, wherein each second blade has a plurality of lobes separated from one another by a recess, extends from a bottom portion to a top portion and has a maximal vertical height along the vertically aligned rotational axis, wherein the rotation of the second blade portion about the vertically aligned rotational axis circumscribes a circle having a radius Ww, and wherein the first blade portion and the second blade portion are fixed with respect to one another, the maximal vertical height for each first blade is at least 1.5 times greater than the maximal vertical height for each second blade, and the radius Ww for the second blade portion is at least 1.5 times greater than the radius Wn for the first blade portion.